Enthalpy and Calorimetry

Exothermic and endothermic reactions, q = mcΔT and calorimetry

Name _____ Class _____ Date _____

USEFUL INFORMATION

$q = mc\Delta T$ · $q_{\text{reaction}} = -q_{\text{solution}}$ · $\Delta H = q_{\text{reaction}}/n$

c(water) = 4.184 J/(g·°C) · c(copper) = 0.385 J/(g·°C) · c(iron) = 0.449 J/(g·°C)

Assume dilute solutions have density 1.00 g/mL and c = 4.184 J/(g·°C) · M(CH_4) = 16.04 g/mol, M(NH_4NO_3) = 80.04 g/mol, M(C_2H_5OH) = 46.07 g/mol

Show units in every step and state whether each ΔH is exothermic or endothermic.

A. Understanding

1. Classify each process as exothermic or endothermic, and give the sign of ΔH: (a) burning wood (b) melting ice (c) dissolving ammonium nitrate in water, which gets cold (d) neutralizing an acid with a base

2. $CH_4\,(g) + 2\,O_2\,(g) \longrightarrow CO_2\,(g) + 2\,H_2O\,(l)$, ΔH = −890.5 kJ. (a) What is ΔH for $2\,CH_4 + 4\,O_2 \longrightarrow 2\,CO_2 + 4\,H_2O$? (b) What is ΔH for the reverse of the original reaction? (c) How much heat is released when 8.02 g of CH_4 burns?

B. q = mcΔT

3. How much heat is needed to warm 40.0 g of copper from 25.0 °C to 75.0 °C?

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

4. A 100. g piece of metal absorbs 900. J and its temperature rises by 20.0 °C. Calculate its specific heat capacity.

5. 5.00 kJ of heat is added to 150. g of water at 22.0 °C. What is the final temperature?

C. Calorimetry

6. 8.00 g of ammonium nitrate is dissolved in 100.0 g of water in a calorimeter. The temperature falls from 22.0 °C to 16.0 °C. Calculate ΔH for dissolving ammonium nitrate, in kJ/mol. (Mass of solution = 108.0 g.)

7. 25.0 mL of 2.00 M HCl and 25.0 mL of 2.00 M NaOH, both at 20.0 °C, are mixed. The temperature rises to 33.5 °C. Calculate ΔH per mole of water formed.

D. Challenge

8. Burning 0.500 g of ethanol ($\Delta H_c = -1366.8$ kJ/mol) under a can containing 200. g of water raises the water temperature by only 12.4 °C. (a) What temperature rise would occur if all the heat went into the water? (b) What percentage of the heat was transferred to the water?

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

9. A 50.0 g iron block at 100.0 °C is dropped into 100. g of water at 20.0 °C in an insulated cup. Find the final temperature. (Heat lost by the iron = heat gained by the water.)

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.